Raghav Bhatia

ch23btech11033@iith.ac.in | raghavbhatia2023@gmail.com | +91 7357181808 | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.81
Children Academy Convent School	Class XII (CBSE)	2023	92.80%
St. Anslem's Sr. Sec. School	Class X (CBSE)	2021	96.20%

WORK EXPERIENCE

Computer Vision Research Intern -- AGC Inc. (AGC Asia Pacific) *Summer 2026 |May - July*

- Evaluated **4 deep learning** approaches -- stereo depth estimation, RGB-D depth completion, 3D Gaussian Splatting, and neural inverse rendering -- on **Python, PyTorch 2.4.0, CUDA 12.1** across **AWS EC2** to solve a core perception bottleneck in company's glass manufacturing robotic grasping line where standard sensors fail on transparent surfaces.
- Reproduced and trained **ClearDepth** in a two-stage pipeline on **EC2 instance** ; matched the paper's target AvgErr of **2.14 px with 2.12 px** using only **1/6th of the paper's compute** -- achieving a **41% accuracy improvement** over the non-pretrained baseline (3.60 px to 2.12 px) via Scene Flow pretraining.
- Reproduced and trained **GAA-TSO** (RGB-D depth completion) from scratch on **EC2**; achieved **87.3% training loss reduction** (0.94 to 0.12) and identified a critical 3D branch collapse where **point cloud loss** contributed **<1% of gradient signal** -- effectively reducing a multimodal network to **2D-only** behaviour.
- Reproduced **3D Gaussian Splatting and neural inverse rendering** pipelines (TSGS & TransparentGS) for geometric reconstruction of transparent objects; TSGS achieved **SSIM of 0.939 and PSNR of 23.18 dB** on complex glassware scenes with **first-surface depth maps** showing sharper transparent boundary separation than **standard plane depth** -- validating their potential for **offline geometric QC** and **optical defect** inspection on company's manufactured glass products.

PROJECTS

CodeWise RAG-Repo Reader |Python | ChromaDB

- Built a **Retrieval-Augmented Generation (RAG)** system in Python that answers natural-language questions over any project codebase with cited, source-grounded responses using **Google Gemini** and **ChromaDB** as the vector store.
- Designed a **hybrid search pipeline** combining semantic vector search and BM25 keyword retrieval, merged via **Reciprocal Rank Fusion** improving answer relevance over single-method retrieval for both meaning-based and exact-term queries.
- Implemented **AST-based code chunking** using Python's ast module to extract functions and classes as complete units, and a **structured LLM output layer** that forces the model to cite sources or admit gaps instead of hallucinating.
- Deployed as a **CLI tool and Gradio web UI**; system indexed 21 chunks across 3 files in under 0.3s , with unit tests passing.

Next Word Predictor using LSTM | Python | NLP

- Built an **NLP-based next-word predictor** using **TensorFlow, Keras, RNN, and LSTM** to learn sequential patterns from text.
- Implemented **text preprocessing, tokenization, vocabulary creation, sequence generation, and padding**.
- Developed an **Embedding-LSTM-Dense Softmax architecture** with **Adam optimizer and categorical cross-entropy loss**, enabling effective multi-class sequence prediction.
- Created a **text-generation pipeline** for sequence inference, probability-based prediction, and iterative next-word generation.

Dynamic Pathfinding System with Multi-Constraint Optimization | C++ | Algorithms

- Designed a tactical A*-based pathfinding simulator with custom cost functions for danger avoidance and cover utilization.
- Implemented graph algorithms and real-time threat assessment for adaptive route optimization.
- Integrated multi-constraint balancing for safe and efficient path selection.
- Developed ASCII visualizations demonstrating algorithmic reasoning and performance outcomes.

SKILLS

Technical Skills

- **Programming Languages** : C , C++ , Python
- **Data Structures and Algorithms:** Proficient, with hands-on problem-solving experience
- **Machine Learning Concepts** : Supervised and Unsupervised ML.
- **Libraries:** NumPy, Pandas, Matplotlib ,TensorFlow, SciKit-learn

Soft Skills : Management , teamwork | Interpersonal Communication | Adaptability, Analytical and Critical Thinking | Patience & Perseverance

RELEVANT COURSES

College Courses

- **Computer Science:** Introduction to Programming ,Introduction to Artificial Intelligence , Computer Vision
- **Mathematics:** Calculus, Linear Algebra, Numerical Methods, Applied Mathematics for Chemical Engineers (including Statistics and Probability)

POSITIONS OF RESPONSIBILITY

Class Representative (CR) | 2024-25

Multimedia Coordinator in Elan & nVision 2025

- Led a 15-member multimedia team to successfully cover and manage content for our annual 3-day college fest.

Volunteer in Office of Career Services(OCS) | 2024-25

- Assisted the placement cell in conducting unbiased and fair exams in internship and placement drive.